

COMPETITIVE INTELLIGENCE — SHORT ANALYSIS

Pentixapharm in CXCR4-Targeted Theranostics

First-in-Class Radiopharmaceutical Platform — CNS Lymphoma, AML, and
Beyond

Prepared for:

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27 May 2026 | Sources: ClinicalTrials.gov, EMA, PubMed, Yahoo Finance, Company Website

Company Snapshot

Attribute	Value
Company	Pentixapharm Holding AG (PTP:GR)
Exchange	Frankfurt Prime Standard (listed 3 Oct 2024)
CEO	Dr. Dirk Pleimes (Group CEO & CMO)
Employees	~35 (Pentixapharm AG)
Locations	Berlin (HQ) + Würzburg
Stock Price (27 May 2026)	€2.27 (XETRA)
52-Week Range	€1.30 – €3.70
Daily Volume (avg)	~3,300 shares (thin liquidity)
Market Cap (est.)	~€50–150M (limited public data — spin-off Oct 2024)
Pipeline	2 Ph3-ready diagnostics + 2 Ph1/2 therapeutics + 1 preclinical
Modality	Theranostics: same ligand for diagnostic (^{68}Ga) and therapy ($^{90}\text{Y}/^{177}\text{Lu}$)

Pipeline Overview

Asset	Indication	Phase	Status
^{68}Ga]Ga-PentixaFor	Hypertension (Prim. Aldosteronism)	Phase 3 (PANDA)	IND submitted to FDA May 2026
^{90}Y]Y-PentixaTher	CNS Lymphoma (PCNSL/SCNSL)	Phase 1/2	RECRUITING (target: n=15)
^{177}Lu]Pentixather	AML (conditioning before HSCT)	Phase 1/2	Dose Level 4 reached (Sept 2025)
^{68}Ga]Ga-PentixaFor	Multiple Myeloma / Lymphoma	Ph3-ready	FDA "Study May Proceed" (Feb 2026)
GT-008	CD24+ solid tumors	Preclinical	First data June 2025

Scientific Foundation

- CXCR4 ligand administered to **>2,000 patients** (diagnostic imaging)
- **150+ publications** on CXCR4 imaging and therapy
- Peer-reviewed dosimetry: ^{177}Lu -Pentixather (J Nucl Med, 2022, PMID: 34413147)
- AML conditioning with Pentixather + TBI (Theranostics, 2025, PMID: 39744224)
- CXCR4 expression validated in >10 tumor indications + cardiovascular + endocrine + inflammatory

Three Key Findings

1. Unoccupied Orphan Space in CNS Lymphoma

No EMA- or FDA-approved therapy exists specifically for PCNSL. Standard of care (HD-MTX-based regimens) is off-label and associated with neurotoxicity. PentixaTher addresses a high-unmet-need orphan space with first-in-class CXCR4 endoradiotherapy. Blood-brain barrier penetration of the PentixaTher ligand has been documented in preclinical and clinical imaging studies — a key advantage over antibody-based approaches in CNS indications.

2. Theranostic Platform = Pipeline-in-a-Pipeline

The CXCR4 ligand scaffold is identical for diagnosis (^{68}Ga) and therapy ($^{90}\text{Y}/^{177}\text{Lu}$). Every new indication where [^{68}Ga]Ga-PentixaFor detects CXCR4 overexpression opens a therapy pathway for PentixaTher. Current: CNS Lymphoma, AML, Multiple Myeloma, Lymphoma, Hypertension — **five indications from one scaffold**. This is a fundamentally different investment profile than single-asset biotechs.

3. Phase 3 Diagnostic De-Risks Therapeutic Pipeline

The PANDA Phase 3 study start in hypertension (IND submitted May 2026) is not just a standalone value driver — it **validates the regulatory platform** with both FDA and EMA. A successful Phase 3 for the diagnostic asset would de-risk the entire CXCR4 pipeline and establish Pentixapharm as a credible radiopharmaceutical developer. Additionally, the diagnostic creates a commercial foothold in clinical practice that prepares the therapeutic rollout.

Near-Term Catalysts

Event	Timeline	Significance
PANDA Ph3 Start (FDA)	H2 2026	First company-sponsored Ph3 — regulatory milestone
AML Pentixather Dose Level 4 Data	2026	Safety & DLT at higher dose levels
CNS Lymphoma Enrollment Complete	~2027	First efficacy signals (ORR at Month 1 & 3)
FDA "Study May Proceed" Myeloma	2026	Second theranostic indication opened

One Watchpoint

Recruitment — CNS Lymphoma: Trial NCT06132737 (target: 15 patients) has been recruiting since November 2023 (~2.5 years). For a Best-of-5 dose escalation in an orphan

CNS indication (incidence ~0.5/100,000), this timeline is consistent with realistic enrollment expectations. The achievement of Dose Level 4 in the parallel AML trial suggests no systemic recruitment issues across the platform.

Sources: ClinicalTrials.gov (NCT06132737), PubMed (PMID: 39744224, 34413147), EMA Human Medicines Register, Pentixapharm Company Website & News (accessed 27 May 2026), Yahoo Finance (PTP.DE).

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